A Physics-Assisted Convolutional Neural Network for Bathymetric Mapping Using ICESat-2 and Sentinel-2 Data

Kaidi Peng, Huan Xie, *Senior Member, IEEE*, Qi Xu, Peiqi Huang, and Ziyi Liu

***Abstract*— Machine learning methods for water depth estimation using remote sensing require accurate prior depth measurements. The successful operation of the Ice, Cloud and land Elevation Satellite-2 (ICESat-2) mission provides depth in shallow water directly; however, its spatial coverage is limited. Machine learning has been used to link optical remote sensing images and ICESat-2 data for bathymetric mapping. Compared with other machine learning models, convolutional neural network (CNN) models utilize the local spatial correlation between adjacent pixels and can, thus, reduce the effect of environmental noise. However, existing CNN and other machine learning models rely on data mining to build a general relationship between water depth and spectral information, and they ignore the known physical law. In this article, we propose a physics-assisted convolutional neural network (PACNN) model. This model incorporates knowledge from radiative transfer theory into a normal CNN model by building a series of spectral feature input layers. In the PACNN model, the spectral information, which is directly related to water depth, is emphasized. Multitemporal ICESat-2 data and Sentinel-2 images were used to validate the model. In experiments with data from three study areas, the PACNN model outperformed the existing CNN model, achieving an accuracy of over 98%. The proposed method can effectively solve the underestimation in deeper water (20–30 m) and reduce the variance of estimates. The superiority of the PACNN model demonstrates how a machine learning model can be assisted by physics theory.**

***Index Terms*— Bathymetric mapping, Ice, Cloud and land Elevation Satellite-2 (ICESat-2), physics-assisted convolutional neural network (PACNN), Sentinel-2, water depth.**

I. Introduction

OCEAN and coastal ecosystems play an important role in maintaining water and carbon cycling [1]. They also provide a physical environment for ocean biodiversity and resource sustainability. As a basic parameter of environmental conditions for coastal regions, high-resolution bathymetric mapping is crucial for coastal utilization and protection. Traditional surveying methods, such as multiple-beam echosounders or airborne LiDAR, allow for accurate water depth retrieval, but are costly. They are also not applicable in sensitive regions where ships and airplanes may be restricted. The development of satellite remote sensing techniques makes large-area satellite-derived bathymetry possible. Passive optical remote sensing can interpret the signal reflected from the seafloor and water column to retrieve water depth, bottom albedo, and inherent optical properties (IOPs). Remote sensing images with different spatio-temporal resolutions (e.g., Landsat 8 [2], [3], WorldView-2 [4], [5], [6], [7], Sentinel-2 [8], [9], [10], [11], GeoEye [12], and Gaofen-1/2 [4]) have been widely used in recent studies. Three categories of methods can be identified for bathymetric mapping in shallow water: physics-based methods, empirical methods, and machine learning methods [13], [14].

Physics-based methods are based on the radiative transfer theory, which describes the physical relationship between water depth, IOPs, bottom albedo, and spectral information. Several studies have developed inversion models and provided the required empirical parameters and equations [15], [16]. By minimizing the error between the measured and modeled spectra, the water depth can be optimized. This type of method does not need prior water depth measurements. However, hyperspectral images are more appropriate than multispectral images as sufficient spectral constraints are needed to uniquely solve the equation. To alleviate the effects of environmental noise and bottom intraclass variability, local maximum likelihood estimation considering the spatial correlation between neighboring pixels has been proposed [17], [18]. Wei et al. [2] applied a physical model to two temporally close optical images and minimized the error between the two sets of modeled and measured spectra. By assuming a constant water depth and bottom albedo during the study period, this method can improve the spectral constraining condition and, thus, achieve higher accuracy. However, in general, physics-based methods are computationally expensive. Incomplete atmospheric correction, water surface reflection, and vertically inhomogeneous IOPs can also introduce great uncertainty into physical models. Research has revealed that chlorophyll concentration, atmospheric scattering, and sun glint can have significant negative effects on inversion accuracy [19], [20].

Empirical methods need prior water depth measurements to build a statistical relationship between the measured spectra

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and depth. Moussavi et al. [7] and Pope et al. [21] built simple single-band regression models, assuming that the backscattering over optically deep water, the effective attenuation coefficient, and the bottom albedo were homogenous over the study area. Other studies have simulated the effects of suspended sediment concentration and bottom reflectance on remote sensing reflectance [22], [23] and have proved that band-ratio models are less sensitive to spatial variability in bottom albedo and IOPs [5], [6], [24], [25]. Smith et al. [26] calibrated a band-ratio model with blue and green bands in WorldView-2 images for supraglacial lake mapping in ablation zones in southwest Greenland. Su et al. [27] proposed a geographically adaptive log-linear model for more complex coastal regions, which is also an effective way to account for heterogeneity in water quality and bottom type. The successful operation of the Ice, Cloud and land Elevation Satellite-2 (ICESat-2) provides high-accuracy depth measurements. Recently, empirical methods combining multitemporal ICESat-2 and Sentinel-2 data have become a hot research topic [8], [9], [10], [11]. Considering the uncertainty caused by environmental and sensor noise, median filtering can be applied to maps produced from multitemporal images [11], [28]. These empirical methods are easy to use, but their performance is highly dependent on the homogeneity of the study area for algorithm calibration. Therefore, they are more suitable for supraglacial lakes and other relatively simple aquatic environments than for coastal regions.

Recently, machine learning models have been applied to water depth estimation. Machine learning has a powerful data mining ability and is more flexible than physics-based methods. It does not need rigorous atmospheric correction and does not have any assumptions about the optical properties of water. Random forest [29], support vector machine [10], convolutional neural networks (CNNs) [4], and other models [14], [30], [31] have all been adopted in recent times. In [4], input to the CNN model includes the neighboring pixels centered on the measured water depth point, including the blue, green, red, and near-infrared bands. Compared with other machine learning models, CNNs can explore the spatial correlation between adjacent pixels, which are likely to share similar water column properties and water depths. Redundant information from neighbors can help to reduce the impact of environmental noise [32]. Even though machine learning methods have a high requirement for training samples, ICESat-2 and other satellite altimeters can be used in place of in situ surveying to calibrate these models.

Existing CNN methods rely on data mining to build the general relationship between the water depth and spectral information, ignoring known radiative transfer theory. Thus, useful features for bathymetric mapping may not be fully learned from the data. Previous research into image classification has proven that incorporating spectral or spatial features in the input layers is an effective way to improve machine learning methods [33], [34], [35], [36]. Therefore, in this article, we propose a physics-assisted convolutional neural network (PACNN) model for bathymetric mapping, where the assisting spectral feature originates from the physics-based model. We aim to take the advantage of prior knowledge from radiative transfer theory to facilitate the training process of the CNN model. In the PACNN model, we maintain the natural logarithm value of each band and construct a spectral feature space for every possible combination of two bands. These feature spaces are transformed into grids as the input layer. The exponential relationship between water depth and spectral information is revealed directly in the PACNN model. By combining multitemporal ICESat-2 LiDAR data and Sentinel-2 multispectral images, we utilized the PACNN model to estimate water depth in three coastal regions, which served to demonstrate its superiority relative to existing techniques.

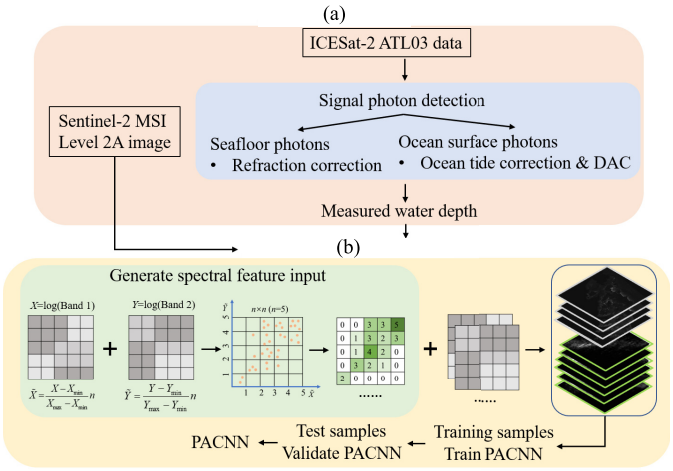

Fig. 1. Flowchart of the proposed PACNN model. (a) Data Preparation. (b) Build the PACNN Model.

II. Method

The proposed PACNN model first constructs training and test samples by calculating the water depth from corrected ICESat-2 measurements and generates input layers for each Sentinel-2 image to train individual models for each image. The calibrated model is then used to estimate the water depth for the corresponding Sentinel-2 image. A brief flowchart for this process is shown in Fig. 1.

A. Data Preparation

1) ICESat-2 ATL03 Data: The ICESat-2 satellite was launched in September 2018 carrying the Advanced Topographic Laser Altimeter System (ATLAS), which produces green (532 nm) laser pulses approximately every 0.7 m along the ground tracks. The standard ATL03 products contain geolocated ellipsoidal height based on WGS84. Three strong laser beams were used in this study. The first step is to separate the seafloor and ocean surface signal photons. We employed the auto-adaptive multilevel seafloor recognition and land sea classification (AMSRLC) method [37] to complete this step. Second, to offset the effect of the water surface refraction, we recalculated the geographical position of each seafloor photon [38].

Third, the measurements of photons are still affected by many geophysical factors [39]. The solid tide, ocean loading,

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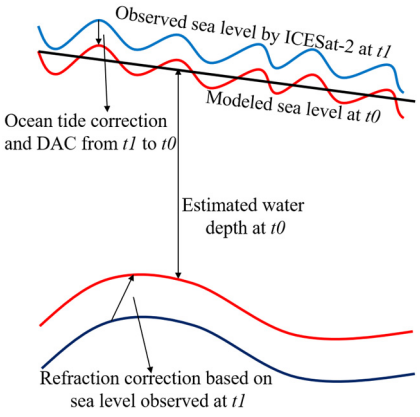

Fig. 2. Water depth estimation from ICESat-2.

and solid Earth pole tide can affect both seafloor and ocean surface simultaneously, thus having little impact on water depth. The magnitude of ocean pole tide is ignorable compared with the ICESat-2 measurement uncertainty. These geophysical corrections, even though have been applied to the ICESat-2 ATL03 product, are not considered in this step. ATL03 provides reference values and models for ocean tide and dynamic atmospheric correction (DAC) without applying them. Their effects cannot be ignored. We calculated the corresponding DACs and ocean tide values at the Sentinel-2 image acquisition times (t0) for each ocean surface photon according to the AVISO 2-D gravity waves model (MOG2D) [40] and Goddard Ocean Tide 4.8 (GOT4.8) [41], respectively [39]. The acquisition time of a certain ocean surface photon is denoted as t1. The ocean tide correction and DAC at t1 are subtracted from the observed height for each ocean surface photon at t1. The two corrections at t0 are then added to obtain the height at t0. After this, a plane is fit using all the corrected ocean surface photons in the study area to model the instantaneous sea level at t0. Finally, the measured water depth at t0 is the height difference between the modeled sea level and the corrected seafloor photons. Fig. 2 depicts water depth retrieval at t0 graphically.

2) Sentinel-2 Multispectral Images: The Sentinel-2A and 2B satellites, which each carry a multispectral instrument (MSI), were launched in June 2015 and March 2017, respectively. The resolution of the imagery ranges from 10 to 60 m, with a five-day revisit period. The Level-2A surface reflectance product processed by the European Space Agency Sen2Cor toolbox was used in this study. Band 1 (coastal blue) with a 20-m resolution was resampled by nearest interpolation to match band 2 (blue), band 3 (green), and band 4 (red) with a 10-m resolution. These four bands were selected to retrieve the bathymetry due to the limited penetration ability of longer wavelengths.

For each Sentinel-2 pixel, the measured points located in the pixel are averaged and assigned to the pixel as its measured water depth. The pixels that have both a measured spectrum and water depth are used as either training or test samples in the PACNN model. The water depths of the other pixels are predicted from the trained model.

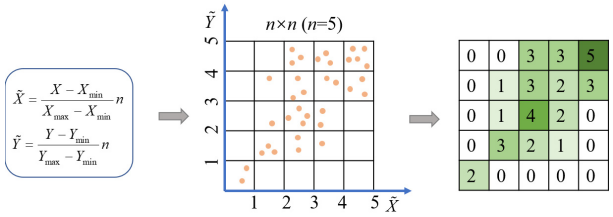

Fig. 3. Generation of the spectral feature input layers. X and Y are the natural logarithm values of the two different bands.

B. Building of the PACNN Model

The regular input of a CNN model is the neighboring pixels centered on the target pixel. In the proposed model, we added a physics-based spectral feature in the input layers to fully exploit spectral information in the training process.

In previous studies, double-band models have been shown to be more accurate than single-band models [5], [6], [24], [25]. The radiative transfer theory for optically shallow water integrating two bands is written as shown in the following equation [22]:

$$\ln\left(\frac{R_1}{R_2}\right) \approx (g_2 - g_1)z + \ln\left(\frac{A_{\mathrm{d}1} - R_{\infty 1}}{A_{\mathrm{d}2} - R_{\infty 2}}\right) \tag{1}$$

where R and z are the measured spectrum and water depth, respectively. R_∞ denotes the backscattering of optically deep water. A_{d} is the bottom reflectance and g is the effective attenuation coefficient. Subscripts 1 and 2 identify two different bands. The assumption of (1) is that the effect of R_∞ on the measured spectrum is negligible, compared with reflectance from the bottom. The prior knowledge from (1) is used to construct the spectral feature input layers.

For every possible combination of two bands, the raw reflectances are converted to their natural logarithm values (i.e., X, Y). To make their coordinate range match the regular local window input, these values are standardized by the corresponding global minimum and maximum values via the following equation:

$$\begin{aligned} \tilde{X} &= \check{X}n = \frac{X - X_{\min}}{X_{\max} - X_{\min}} n \\ \tilde{Y} &= \check{Y}n = \frac{Y - Y_{\min}}{Y_{\max} - Y_{\min}} n \end{aligned} \tag{2}$$

where $X_{\min}$ and $Y_{\min}$ are the global minimum values of X and Y, respectively. $X_{\max}$ and $Y_{\max}$ are the global maximum values of X and Y, respectively. $\check{X}$ and $\check{Y}$ are then scaled by being multiplied by the window size n to obtain $\tilde{X}$ and $\tilde{Y}$. The spectral space formed by $\tilde{X}$ and $\tilde{Y}$ is transformed into a raster representation by marking these points in the Cartesian coordinate system and counting the number of points located in each grid cell. In this way, a spectral feature input sharing the same size as the regular input is generated. Fig. 3 illustrates this process in brief. The regular input and the new spectral feature input are combined to form the input of the PACNN model. The former is maintained to preserve the data fidelity, as the latter may cause information loss in the transformation.

The structure of the PACNN model is presented in Fig. 4. Two convolutional layers and two activation layers are

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TABLE I
DATA COLLECTION FOR THE THREE EXPERIMENTS

Location	Culebra Island	Yongle Atoll	Discovery Reef
Sentinel-2	15 Jan. 2019	24 Feb. 2019	20 Jan. 2019
	20 Jan. 2019	26 Sep. 2020	15 Jan. 2020
	20 Mar. 2020	21 Oct. 2021	18 Feb. 2021
	28 Feb. 2021	15 Dec. 2021	5 Mar. 2021
ICESat-2	20 Oct. 2018 and 24 Oct. 2018	22 Oct. 2018 and 22 Feb. 2019	22 Oct. 2018 and 22 Feb. 2019
	19 Jan. 2019 and 20 Apr. 2019	21 Apr. 2019 and 24 May 2019	23 Mar. 2019 and 21 Apr. 2019
	17 Jan. 2020 and 21 Jan. 2020	20 Oct. 2019 and 18 Nov. 2019	24 May 2019 and 19 Aug. 2019
	21 Apr. 2020 and 17 Jul. 2020	22 Nov. 2019 and 19 Jan. 2020	20 Oct. 2019 and 18 Nov. 2019
	21 Jul. 2020 and 16 Oct. 2020	20 Feb. 2020 and 20 Mar. 2020	22 Nov. 2019 and 19 Jan. 2020
	15 Jan. 2021 and 15 Jul. 2021	19 Apr. 2020 and 19 Jul. 2020	20 Feb. 2020 and 20 Mar. 2020
		20 Aug. 2020 and 19 Nov. 2020	19 Apr. 2020 and 19 Jul. 2020
		17 Apr. 2021 and 16 May 2021	20 Aug. 2020 and 19 Nov. 2020
		19 May 2021 and 17 Jul. 2021	19 Mar. 2021 and 17 Apr. 2021
		18 Aug. 2021	16 May 2021 and 19 May 2021
			17 Jul. 2021 and 18 Aug. 2021

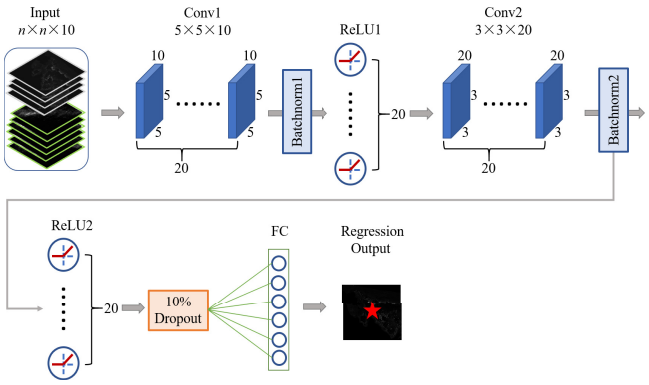

Fig. 4. Structure of the PACNN model. n is the input window size. The red star is in the center of the local window. Conv: convolutional layer, Batchnorm: batch normalization layer, rectified linear unit activation function (ReLU): activation layer using ReLU function, Dropout: dropout layer, and FC: fully connected layer.

employed to operate the regression. The number of neurons in the convolutional layers is 20. A 10% dropout layer is also added after the convolutional layers to prevent the network from overfitting. A fully connected layer and a regression layer linking the water depth and the trained neurons are placed at the end of the network. The network optimizes the half-mean-squared-error loss for this general regression task. The MATLAB deep learning toolbox was used in the experiments conducted in this study.

The whole network is trained for 100 epochs. The size of each subset is 128. For each iteration, 128 training samples are sequentially selected to train the model. One epoch finishes when all the training samples are trained in turn. The training samples are used to evaluate the gradient of the loss function and update the weights. The initial learning rate is set to 0.001, and the learning rate is lowered after 50 epochs. The only user-defined parameter in the PACNN model is the neighboring window size n. The R^2, root-mean-square error (RMSE), and mean absolute error (MAE) values of the training and test samples are used in this article to evaluate the performance of the PACNN model and make a comparison with the existing CNN model and one empirical model.

III. DATA AND EXPERIMENTS

A. Experimental Data

The three study areas are presented in Fig. 5(a1)–(c1). The first study area is Culebra Island located in Puerto Rico. The size of Culebra Island is about 10 × 15 km. Yongle Atoll and Discovery Reef are in the Xisha Archipelago in the South China Sea. The former is about 18 × 23 km, and the latter is slightly smaller, at about 13 × 27 km. Yongle Atoll consists of peripheral reefs and a central lagoon. It is covered by coral, rocks, rubble, and sand [42]. To the south of Yongle Atoll is Discovery Reef, where the depth of the lagoon reaches up to 25 m. The water areas were masked using the normalized difference water index (NDWI) [43] plus Otsu's thresholding method and manual correction. The Sentinel-2 and ICESat-2 data utilized for each region are listed in Table I. The locations of all the used ICESat-2 measurements are marked in Fig. 5(a1)–(c1).

B. Demonstration of the Spectral Feature Input

The exponential attenuation of reflectance with water depth is illustrated in Fig. 6(c). This is the basis of water depth inversion from optical remote sensing images. A band with a shorter wavelength can penetrate deeper water while a band with a longer wavelength is more sensitive to smaller variations in depth in shallow water. The rationale of our physics-based spectral feature input is also demonstrated graphically. The distribution of the natural logarithm values of the green and blue bands for the whole study area is shown in Fig. 6(a), where an obvious clustered distribution for pixels with similar water depths can be observed. Furthermore, for the deeper water, the deviation between ln(green) and ln(blue) is larger. This information is also beneficial for predicting water depth, in addition to the lower reflectance in deeper water. It is an advantage of using two bands.

A comparison to the scatterplots for the original green and blue bands is provided in Fig. 6(b) to illustrate the need for the natural logarithm values, where the distribution is heavily unbalanced and indistinguishable, due to the exponential trend.

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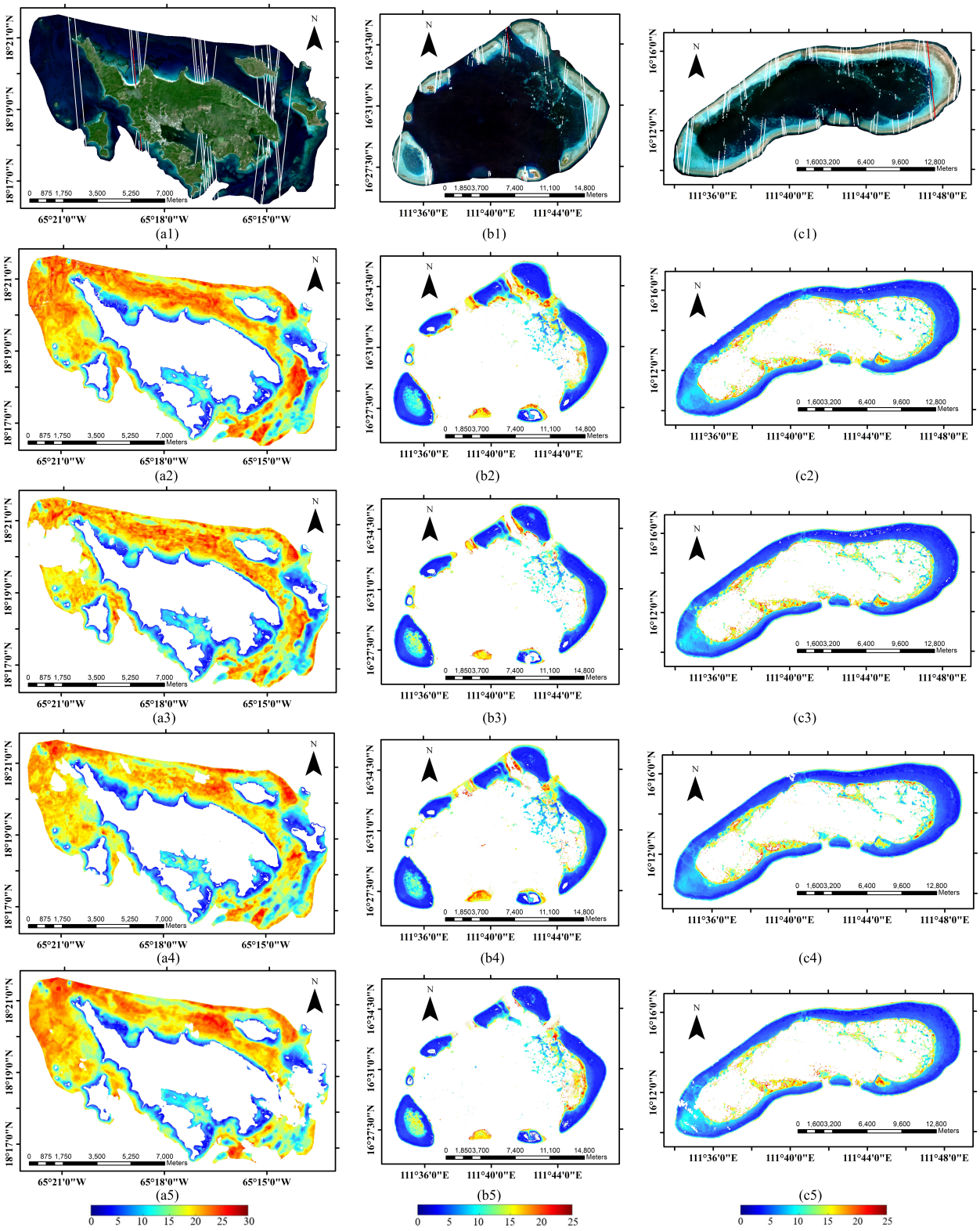

Fig. 5. Three study areas and the generated bathymetric maps: (a1) Culebra Island image acquired by Sentinel-2A on January 15, 2019; (b1) Yongle Atoll image acquired by Sentinel-2A on February 24, 2019; and (c1) Discovery Reef image acquired by Sentinel-2B on January 15, 2020. The white lines are the collected ICESat-2 tracks. The red line is the track used for display in Fig. 10. Bathymetric maps derived from the PACNN model for: (a2)–(a5) Culebra Island; (b2)–(b5) Yongle Atoll; and (c2)–(c5) Discovery Reef using the multitemporal ICESat-2 and Sentinel-2 data. The projection of Culebra Island is WGS_1984_UTM_Zone_19N. The projection of Yongle Atoll and Discovery Reef is WGS_1984_UTM_Zone_49N.

Therefore, the prior information from the physics-based model is introduced to better exploit the relationship between the water depth and spectral information.

C. PACNN Model Parameters

The only parameter in the PACNN model that must be specified is the window size. This also controls the scaling

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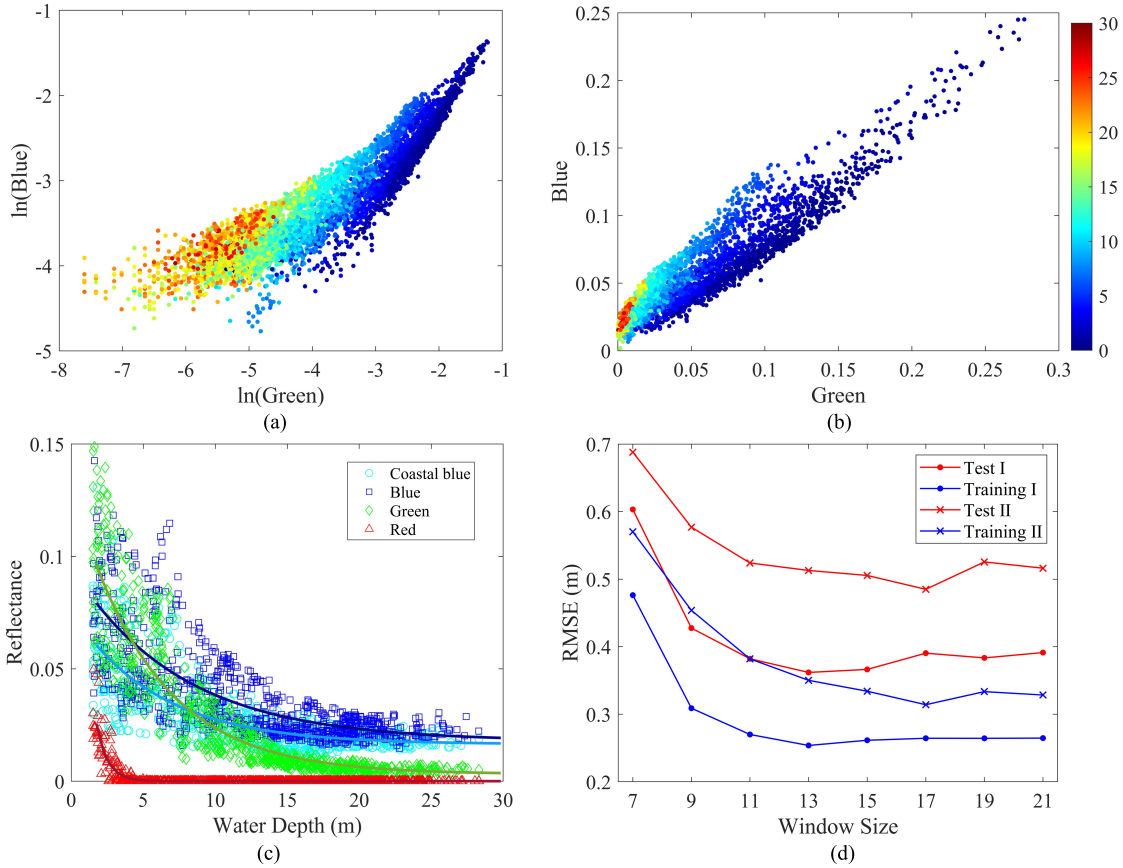

Fig. 6. (a) Scatterplots of ln(green) and ln(blue). (b) Scatterplots of the original green and blue bands. The color of the points in (a) and (b) represents the water depth (see the color bar in the right). (c) Reflectance changes with water depth, where exponential fitting is shown for each band. (d) Different RMSEs under different window sizes from 7 to 21. The optimal window size for Yongle Atoll (I) is 13, and for Discovery Reef (II), it is 17.

factor (i.e., n) in the construction of the spectral feature input. A larger window size means that the spatial association is preserved in a larger region and the differences in the spectral feature are magnified. An optimal window size should be able to avoid the negative effects of spatial variation and observation noise. We assessed the accuracy of the PACNN model with different window sizes ranging from 7 to 21. The RMSE values for Yongle Atoll and Discovery Reef under different window sizes are provided in Fig. 6(d). We chose 13 as the optimal value for Yongle Atoll as a larger window size can cause a high computational load and slightly higher error. Similarly, 15 and 17 were set as the optimal window sizes for Culebra Island and Discovery Reef, respectively.

D. Comparison With the Original CNN Model

For the three regions, we trained the CNN and PACNN models with the same structure as Fig. 4 and made a comparison between them. The workflow shown in Fig. 1 was applied to the ICESat-2 and Sentinel-2 data. The generated maps for the three regions are presented in Fig. 5, which are in good agreement with each other. No significant bias appears between different times.

Figs. 7 and 8 show the comparison of the training and test errors, respectively. Overall, the proposed PACNN model can avoid the underestimation that occurs for the CNN model in deeper water (20–30 m), which can be seen clearly in Figs. 7(b) and (c) and 8(b) and (c). The variance in the water depth estimation of the PACNN model is significantly lower than that of the CNN model. The decrease of RMSE from the PACNN to CNN models is over 47.5 cm for the training samples at Culebra Island. At Yongle Atoll, the accuracy of the PACNN model is more than 99% for both the training and test samples, which is much better than the CNN model. The RMSE decreases by over 40 cm from the CNN to PACNN models for the training samples. For the test samples shown in Fig. 8, the decrease of RMSE is also over 35 cm for both regions.

The accuracy in different water depth intervals is compared in Fig. 9. The error produced from the PACNN model is no more than 1 m over almost the whole region. The PACNN and CNN models show similar errors in water depths of less than 5 m, but a systematic error appears in the deeper water for the CNN model. At Yongle Atoll, the CNN model overestimates the water depth between 5 and 15 m and underestimates the water depth over 15 m. In Fig. 9(b), obvious underestimation can be observed for the water depth over 5 m, and the error reaches 3 m when the water depth is over 20 m. All in all, the PACNN model can grasp the spectral feature in different depths, resulting in a significant improvement in the model fitting.

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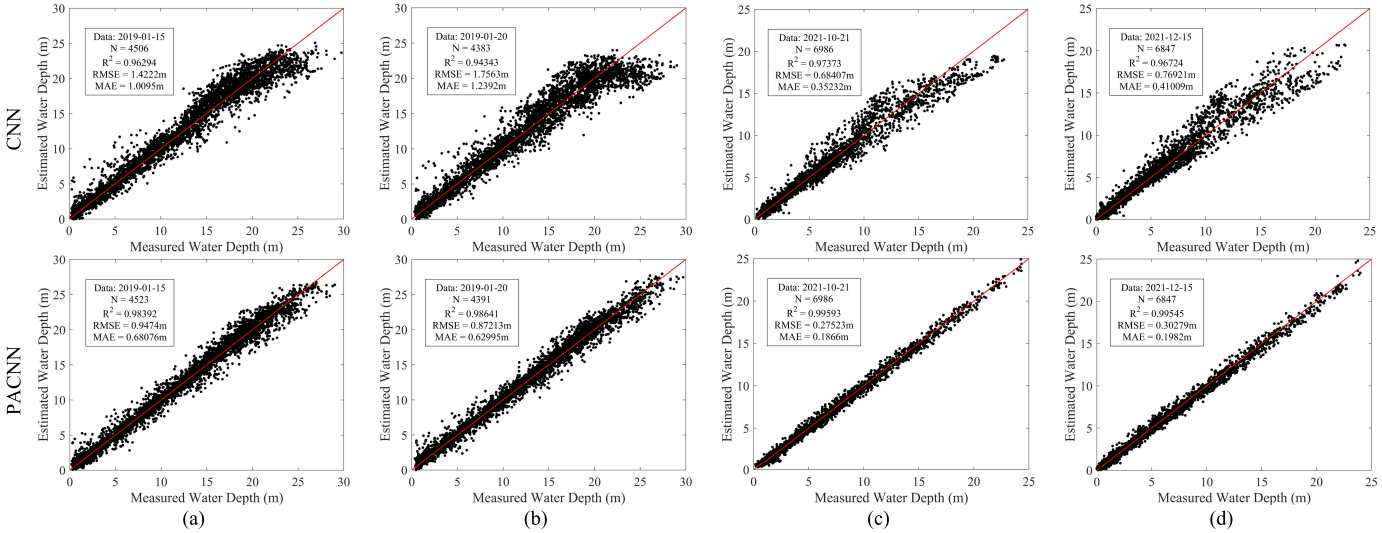

Fig. 7. Training comparison between the CNN and PACNN models: (a) and (b) are scatterplots of Culebra Island on January 15, 2019 and January 20, 2019, respectively, and (c) and (d) are scatterplots of Yongle Atoll on October 21, 2021 and December 15, 2021, respectively. The red line is the 1:1 line. N is the number of used training samples. Absolute errors larger than the 99th percentile and negative values were regarded as outliers and removed. No more than 50 training samples were deleted for Culebra Island. No more than 70 training samples were deleted for Yongle Atoll.

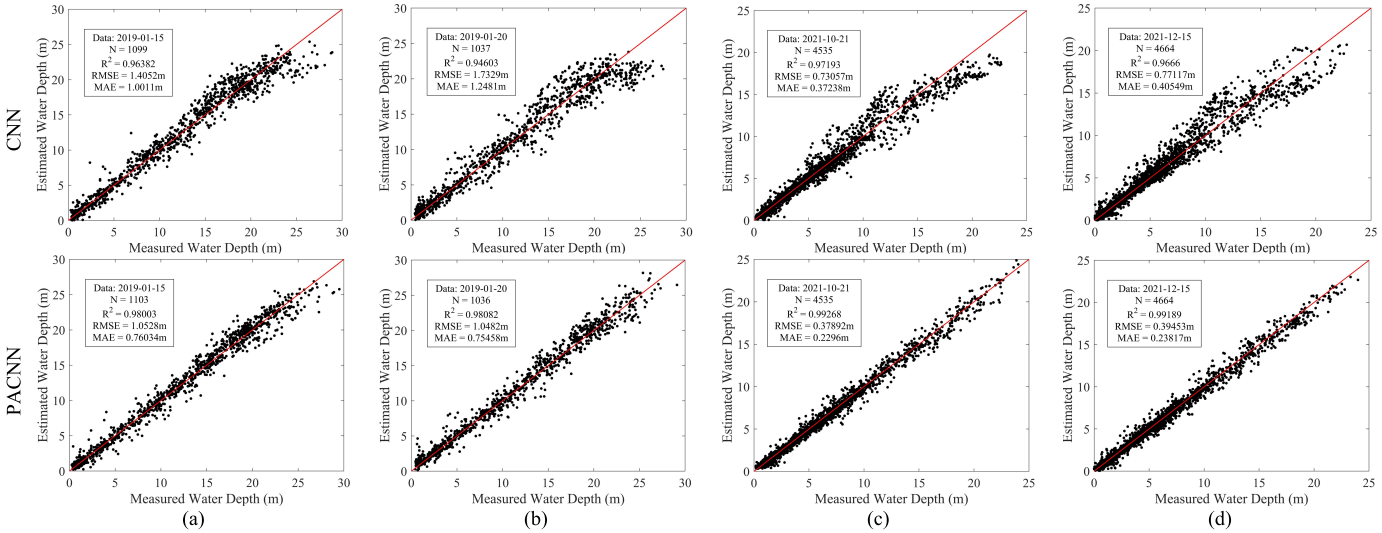

Fig. 8. Test comparison between the CNN and PACNN models: (a) and (b) are scatterplots of Culebra Island on January 15, 2019 and January 20, 2019, respectively, and (c) and (d) are scatterplots of Yongle Atoll on October 21, 2021 and December 15, 2021, respectively. The red line is the 1:1 line. N is the number of used test samples. Absolute errors larger than the 99th percentile and negative values were regarded as outliers and removed. No more than 15 test samples were deleted for Culebra Island. No more than 50 test samples were deleted for Yongle Atoll.

In addition to the statistical results, we also present the estimated water depth from the CNN and PACNN models for a specific track over each study area. Similar conclusions can be made by inspecting Fig. 10, where the PACNN model is closer to the actual measurements. At Culebra Island, the CNN model underestimates the water depth over 14 m, where the error of the CNN model is at least 2 m larger than that of the PACNN model. For the second row, in the first 500 m along the track, the CNN model has trouble estimating the water depth of around 20 m, but the prediction from the PACNN model has an error of less than 1 m. The CNN model also overestimates the water depth below 10 m, causing an error of about 1 m. Meanwhile, the PACNN model is very accurate in this shallow water. The overestimation of water depths of less than 10 m is also obvious in the CNN model, compared with the PACNN model in the third row.

In addition, we compared the training processes of the CNN and PACNN models in Fig. 11. The decrease of RMSE as the iterations proceed for the PACNN model is more significant than for the CNN model, especially in the first few iterations. The PACNN model can exploit more useful information from the spectral feature input and has a better fitting ability. For each region, similar iterations are needed for the CNN and PACNN models to optimize the models. In total, 4000 and 3000 iterations are adequate for Discovery Reef and Yongle Atoll, respectively. Furthermore, the training efficiency of the PACNN and CNN models is presented in Table II, including seconds needed for training the model and the number of

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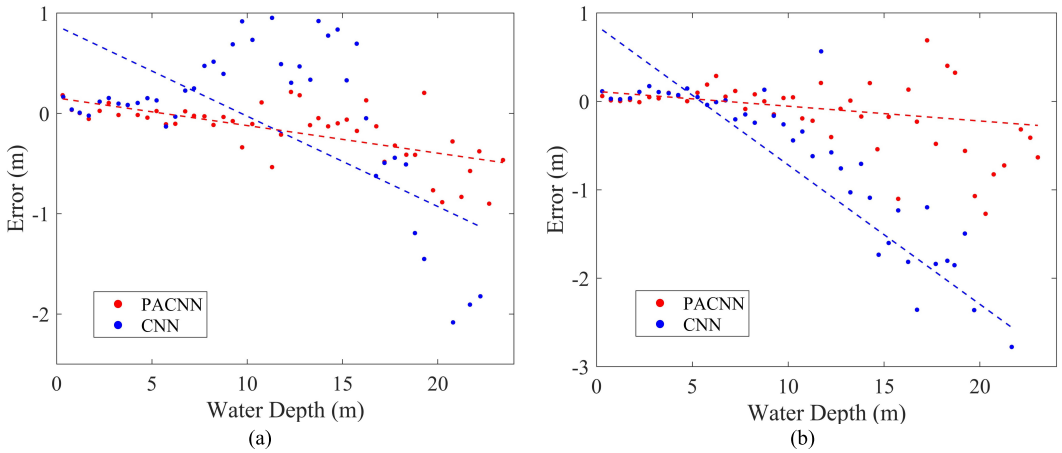

Fig. 9. Comparison of errors between the PACNN and CNN models in different water depth intervals: (a) analysis for Yongle Atoll on February 24, 2019 and (b) analysis for Discovery Reef on January 15, 2020. The error equals estimated water depth minus measured water depth. Each point represents an average of an interval of 0.5 m. The dashed line is the linear fitting for the CNN or PACNN models. Absolute errors larger than the 99th percentile and negative values were regarded as outliers and removed.

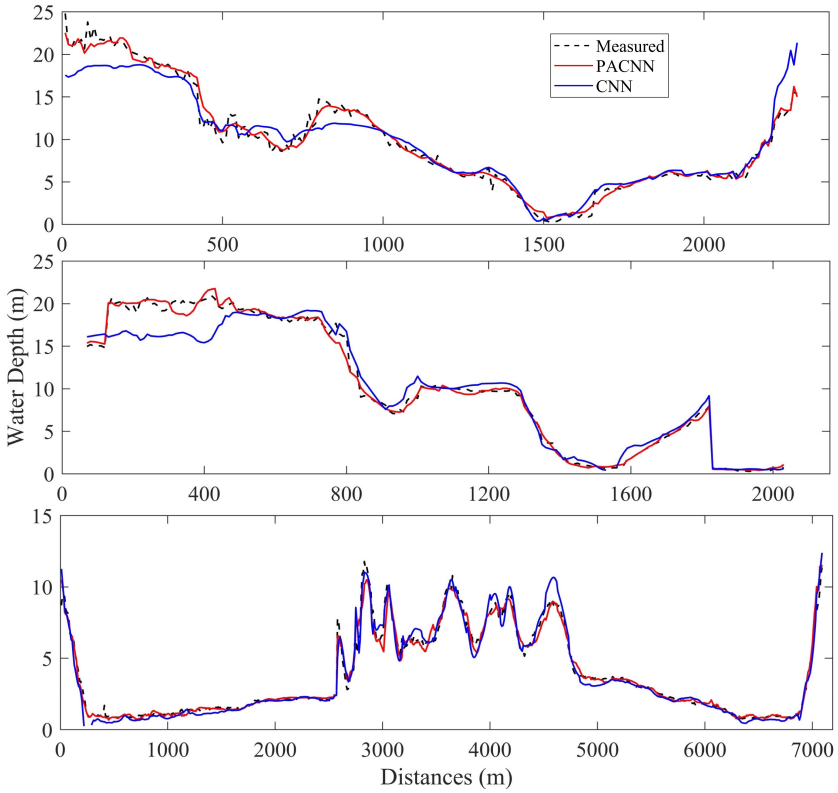

Fig. 10. Profiles of the measured water depths and the PACNN and CNN model estimates. The first row is Culebra Island on March 20, 2020. The second row is Yongle Atoll on February 24, 2019. The third row is Discovery Reef on February 18, 2021. The displayed tracks are the ones marked in red in Fig. 5.

training samples. All the experiments were carried out using MATLAB (R2020a) on a laptop with an 11th Gen Intel[1] Core[1] i5-11300H CPU at 3.10 GHz. In general, a slightly longer time is needed for the PACNN model. The spectral feature input is quite different from the regular input, so the PACNN model needs more time to learn from these additional layers. The use of more training samples also requires more time. For example, around 10% additional time was needed for nearly 7000 training samples, but almost 20% more time was consumed when the number of training samples approaches 10 000.

E. Comparison With Quadratic OBRA

We made a comparison between PACNN and the quadratic optimal band ratio analysis (OBRA) model, which is a well-established empirical method [6]. The green and blue bands were selected to generate quadratic OBRA. Fig. 12 presents the results of quadratic OBRA using the same images as Figs. 7 and 8. R^2 of PACNN is at least 9% larger than that of OBRA. RMSE is at least 1.1 m lower. In addition, it is

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TABLE II
TRAINING EFFICIENCY OF THE CNN AND PACNN MODELS

Culebra Island			Yongle Atoll			Discovery Reef		
Seconds for CNN (s)	Seconds for PACNN (s)	Training samples	Seconds for CNN (s)	Seconds for PACNN (s)	Training samples	Seconds for CNN (s)	Seconds for PACNN (s)	Training Samples
54	54	4435	81	93	7061	134	162	9600
52	56	4570	88	94	6948	138	166	9925
46	52	4347	77	87	6940	141	173	9989
49	53	4236	82	90	6933	140	166	9945

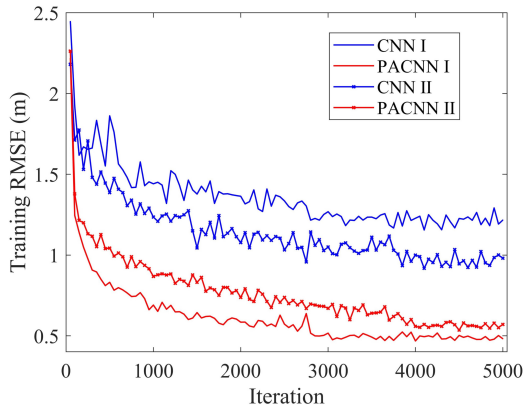

Fig. 11. Error curves for the model training for Discovery Reef (I) on March 5, 2021 and Yongle Atoll (II) on December 15, 2021. Training RMSEs for iterations over 5000 are not shown, as the training processes were stable before 5000.

worth mentioning that 1.5%–3.5% of estimations are negative in Culebra Island and 7%–10% are negative in Yongle Atoll. Many more invalid estimations were generated in OBRA than that in PACNN. OBRA and PACNN both use natural logarithm values of reflectance as input, but the former builds an explicit function using two bands, while the latter utilizes machine learning with multiple bands.

F. Capability of the PACNN Model

Accurate water depth inversion from optical remote sensing data is crucial for the wider application of such data. We calculated the maximum detectable water depth and relative accuracy for the PACNN model to evaluate its capability to estimate a broad range of depths [44], [45]. First, we drew a curve between the truncated water depth and the corresponding test R^2, where the truncated water depth ranged from 10 to 25 m. Only the samples shallower than the truncated water depth were used in training the PACNN model to obtain the corresponding test R^2. The results for Culebra Island and Yongle Atoll are presented in Fig. 13(a) and (b), respectively.

The accuracy is lowest for the shallowest water, as only a limited range of depths was used for the calibration. The trend of the reflectance is overwhelmed by the variation caused by bottom variability and environmental and sensor noise. When the water depth becomes deeper, the exponential law becomes gradually dominant, achieving a higher R^2. In Fig. 13(a), the increase of accuracy becomes marginal when the water depth is over 20 m. One possible reason for this effect is that the reflectance changes by a smaller amount as depth increases for water depth over 20 m. The received signal from water only includes backscattering from within the water column. No reflectance from water bottom can be received. This point can be regarded as the maximum detectable water depth for Culebra Island. In Fig. 13(b), the accuracy increases steadily with the truncated water depth. The tested water depth is, thus, detectable for Yongle Atoll. This result indicates that a sufficient range of depths is needed to build a reliable model. Meanwhile, deeper water than the maximum detectable water depth can disturb the model and cause a gradual drop in accuracy. Even though signals from deeper water can be detected, they cannot provide valid information for the water depth retrieval. These conclusions are in a good agreement with previous bathymetric mapping studies using OBRA [44], [45]. The inherent physics-based limitations of optical remote sensing restrict the capability of PACNN. The maximum detectable water depth is different in these two regions, as the value depends on the sensor, water type, and imaging environment. According to Legleiter and Overstreet [5], a finer radiometric resolution can provide greater depth retrieval precision and a greater maximum detectable depth.

The uncertainty of the estimations is presented in Fig. 13(a) and (b) for Culebra Island and Yongle Atoll, respectively. The uncertainty is defined as the absolute value of the error divided by the measured water depth. The uncertainty is relatively high in shallower water and decreases gradually with the water depth. This is likely because of the decrease of variance in the measured reflectance (see Fig. 6). The decrease becomes marginal when the water depth is over 20 m. One reason for this is that the water depth exceeds the maximum detectable water depth. Another is that the limited training samples in this interval lead to bias. In both regions, the 10–25-m water depth has the highest relative accuracy, at around 5% uncertainty for Culebra Island and 2% for Yongle Atoll.

IV. DISCUSSION

A. Superiority of the Proposed PACNN Model

The proposed PACNN model greatly improves the accuracy of satellite-based bathymetric mapping. ICESat-2 has a 91-day repeat cycle, with a high along-track density of photons (about 70-cm interval). It provides an unprecedented opportunity for global height measurement. In our experiments, geophysical corrections were applied to collected ICESat-2 measurements, considering the variability in ocean surface and seafloor. This operation can make full use of available water depth measurements and address one of the main limitations of using machine learning. Our experiments in three coastal regions

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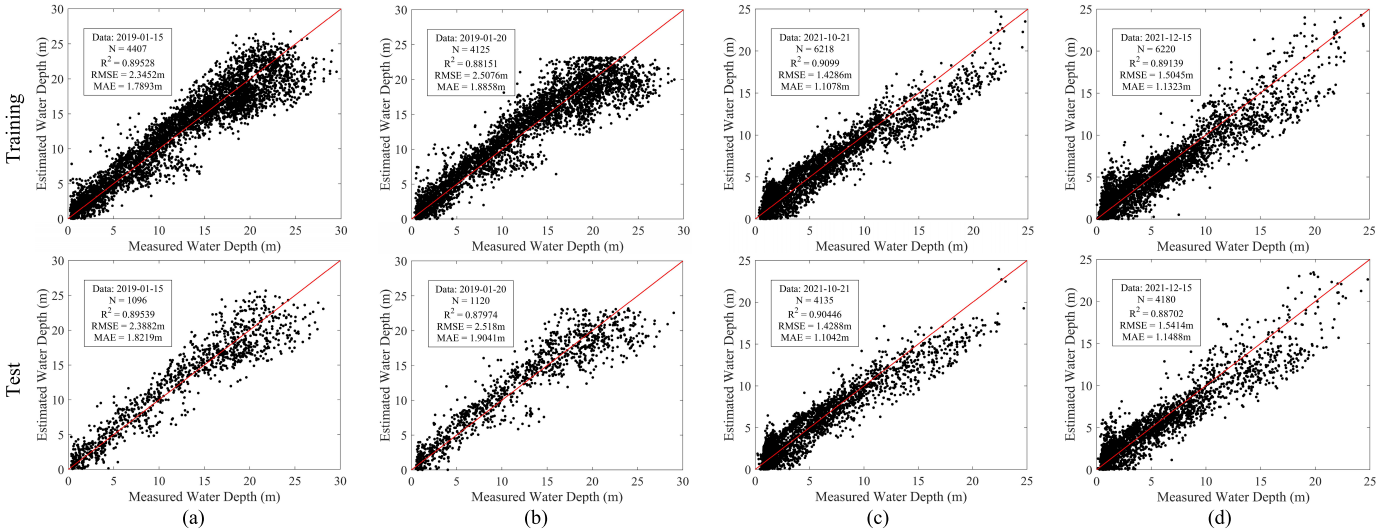

Fig. 12. Accuracy of quadratic OBRA for training and test samples: (a) and (b) are scatterplots of Culebra Island on January 15, 2019 and January 20, 2019, respectively, and (c) and (d) are scatterplots of Yongle Atoll on October 21, 2021 and December 15, 2021, respectively. The first row shows the results of training samples. The second row shows the results of test samples. The red line is the 1:1 line. N is the number of used samples. Absolute errors larger than the 99th percentile and negative values were regarded as outliers and removed. About 200 training samples and 50 test samples were deleted for Culebra Island. About 700 training samples and 500 test samples were deleted for Yongle Atoll.

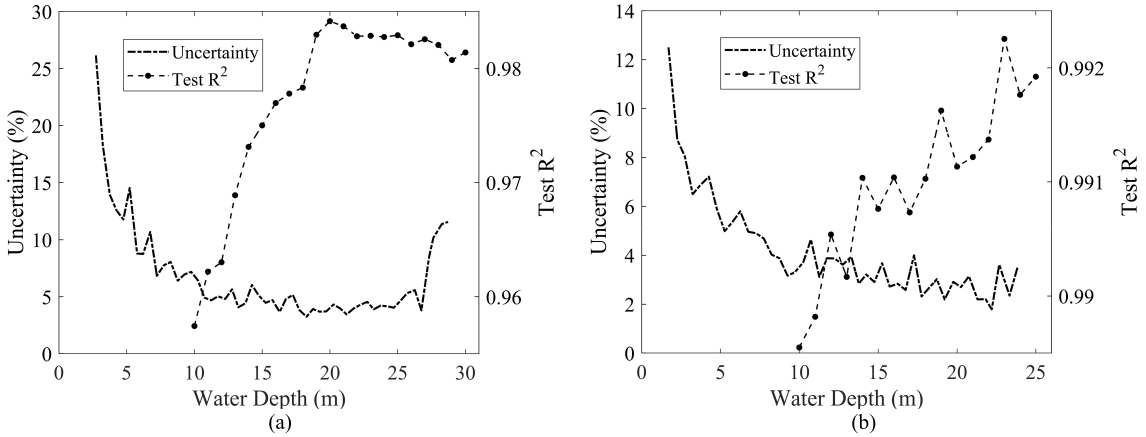

Fig. 13. Test R^2 changes with truncated water depth. The step of the water depth is 1 m. Uncertainty (%) changes with water depth. The step of the water depth is 0.5 m: (a) results for Culebra Island on February 28, 2021, where 80% of the data were used for the training and 20% of the data were used for the test and (b) results for Yongle Atoll on December 15, 2021, where 60% of the data were used for the training and 40% of the data were used for the test.

demonstrate the suitability of PACNN. PACNN can improve the RMSE of bathymetry using CNN by 30–50 cm and RMSE of quadratic OBRA by at least 1.1 m. The absolute error of PACNN is less than 1 m in the range of 30-m water depth. The maximum detectable depth is over 20 m, and the decrease in accuracy in deeper water is slight. These capabilities of PACNN make the technique suitable for water depth inversion in coastal regions.

We further demonstrated the superiority of PACNN by making a comparison with three types of methods. The R^2 of the linear method used in coastal regions in [8] is no more than 97%. In Yongle Atoll, the RMSE is 0.6 m higher than that of PACNN. For machine learning methods, RMSE is improved from 2.5% to 7% of the range of depth to at most 3.5% compared with CNN in [4] and local back propagation neural network (BPNN) in [27]. The physics-based method in [46] reaches an accuracy of no more than 94% in a range of 15 m, and the bias is over 1.1 m for 10–15 m depth. PACNN is superior to these methods in coastal regions as it is more flexible to cope with the heterogeneity in water quality and bottom type.

The PACNN model is an example of a combination of physics-based and CNN models. Quantitative remote sensing algorithms integrating physical modeling and machine learning have been summarized in [47] and [48]. In general, machine learning methods are known to ignore the laws of physics, while physical methods are rather complex and difficult to apply in the real world. Here, we adopt an exponential law from the radiative transfer theory to construct spectral feature input layers in the PACNN model. This prior information from the laws of physics benefits model development by maximally mining the information from the training samples. In other words, with the natural logarithm of the spectra as the input, we replace the computationally expensive radiative transfer model with a flexible CNN model. The interpretability of the laws of physics and the powerful learning capability of a CNN

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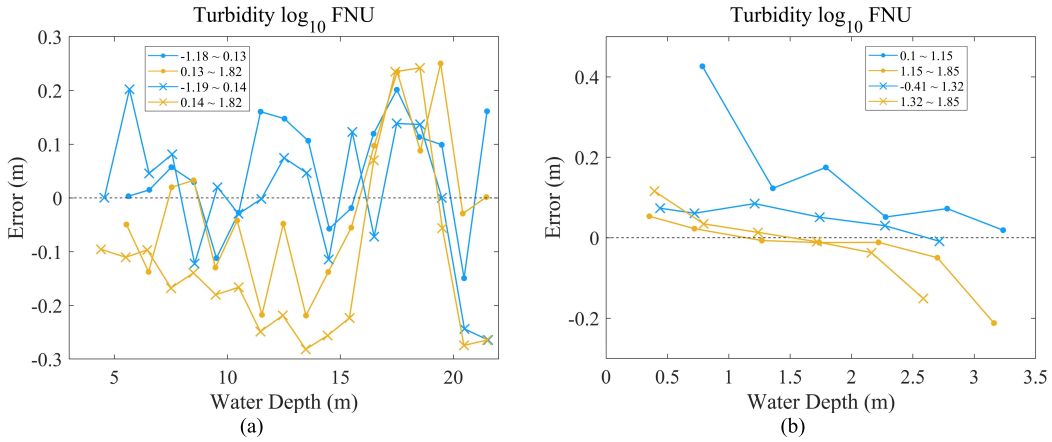

Fig. 14. Impact of turbidity on water depth retrieval accuracy: (a) results for Culebra Island on January 15, 2019 (lines with dots) and January 20, 2019 (lines with crosses) and (b) results for Discovery Reef on January 15, 2020 (lines with dots) and February 18, 2021 (lines with crosses). The yellow lines represent water columns with a higher turbidity. The blue lines represent water columns with a lower turbidity. The unit of the labeled numbers is $\log_{10}$ (FNU). The error equals the estimated minus measured water depth.

are combined to improve the estimation accuracy. In the results obtained in this study, the PACNN model showed a superior ability in fitting the relationship between water depth and spectral information, compared to the existing CNN model.

B. Impact of Water Turbidity

The IOPs are the primary controls on retrieval accuracy. Caballero et al. [49] revealed that high turbidity and chlorophyll concentration limit depth retrieval for the physics-based methods. In this study, we evaluated the effect of turbidity on the PACNN model. Sentinel-2 Level 1C products were processed using ACOLITE software version 20211124.0 to obtain the turbidity index [50]. Turbidity [Formazin Nephelometric Units (FNU)] calculation in ACOLITE uses the algorithm of Nechad et al. [51]. The trend of the error with water depth for different turbidities is presented in Fig. 14, where the turbidity is divided into two categories, i.e., higher (i.e., yellow line) and lower (i.e., blue line). The number of samples in each category is the same. Water depths from 4 to 20 m are presented for Culebra Island, while depths of 0–3 m are presented for Discovery Reef. From Fig. 14, it can be seen that the water columns with higher turbidity tend to predict lower water depths. This is likely because the higher turbidity causes stronger backscattering. However, less turbid water may not always result in a higher accuracy with the PACNN model.

C. Generality of the Proposed PACNN Model

We evaluated the generality of the PACNN model by using one region as training data and another region as test data. The accuracy is shown in Table III. Models trained by other regions achieve relatively lower R^2 compared with individual models, as the water quality and imaging conditions are different for each image. However, considering the situation of lacking water depth measurements, the accuracy is comparable with physics-based methods [46]. Therefore, the generality of PACNN makes it an alternative to physics-based methods with lower complexity.

TABLE III

R^2 OF MODELS TRAINED BY ANOTHER REGION TO DEMONSTRATE THE GENERALITY OF PACNN

		Training Data from Yongle Atoll	
		26 Sep. 2020	24 Feb. 2019
Test Data from Discovery Reef	5 Mar. 2021	0.9102	0.8176
	18 Feb. 2021	0.8859	0.8890
	15 Jan. 2020	0.8597	0.8986
	20 Jan. 2019	0.8612	0.9057

D. Limitation of the Proposed PACNN Model

As in many empirical methods, we assume that the backscattering of optically deep water (i.e., R_∞) is negligibly small, compared with the reflectance coming back from the water bottom. Usually, R_∞ varies over the study area. This causes higher variability in the reflectance for a specific water depth. However, it is extremely difficult to determine R_∞ locally and accurately using physics-based methods. The determination of R_∞ also likely introduces more uncertainty in the model. For the PACNN model, this problem is not worth addressing since the model can achieve an accuracy of over 98%, even in the presence of varying R_∞.

Uncertainty may also be introduced in the data processing. First, considering the difference in the data acquisition times, the correction of the water depth measurements relies on the accuracy of the used models (i.e., ocean tide and DAC) and the other unmodeled factors that affect instantaneous sea levels, such as storms and tropical cyclones. Second, the transformation between raster and vector may cause registration errors. Geolocational uncertainty can also be introduced from the WGS84 to UTM projection. Third, in geolocational correction for seafloor photons, we only consider the incident angle of the beams and assume that the local ocean surface is a plane. A solution to this problem can be found in [52], which also considers the slope of waves by the Joint North Sea Wave Observation Project (JONSWAP) model. In the experiments conducted in this study, no significant bias between different

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times was observed, which means that these uncertainties are not very serious.

In the experiments, the spatial resolution of the Sentinel-2 images was 10 m, and averaged depth measurements were assigned to each pixel. Remote sensing images with a finer spatial resolution could be explored in future research. On the one hand, this could reveal the detailed texture in more precise bathymetry. Compared to coarse pixels, abrupt changes at the pixel scale are rare. On the other hand, heterogeneous bottom and water surface roughness can be averaged over the scale of a pixel. However, the variability is likely to be more obvious between pixels with a higher spatial resolution. The sensitivity of the PACNN model to the spatial resolution could be further evaluated using different-resolution images.

Only coastal regions were considered in the experiments conducted in this study. More complex water types such as inland water bodies could be further tested to assess the robustness of the PACNN model. In addition, the consistency between mosaicked images for large-area bathymetric mapping could also be further investigated.

V. Conclusion

In this article, we have proposed the PACNN model for bathymetric mapping, which combines the prior information in physics-based and CNN models to highlight the relationship between spectral information and water depth. The spectral feature input layers allow for more complete exploitation of the useful information in the training samples. Multitemporal ICESat-2 and Sentinel-2 datasets were utilized to estimate the water depth in coastal regions. Four-date Sentinel-2 images in three coastal regions were used to demonstrate the consistency and superiority of the water depth maps produced by the PACNN model. The accuracies obtained in all the experiments were over 98%. In the systematic comparison between the PACNN and CNN models, we found that the PACNN model can solve the problem of underestimation in deep water and can obtain a higher accuracy in all depth intervals. The generality of PACNN makes it an alternative to physics-based methods with lower complexity when local water depth measurements are not available. The PACNN model is a good example of how physical and data-driven models can be combined in quantitative remote sensing. The application of the PACNN model to inland water bodies, multiscale images, and mosaicked images should be further studied in the future.

References

[1] S. A. Kulp and B. H. Strauss, "New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding," *Nature Commun.*, vol. 10, no. 1, p. 4844, Dec. 2019.

[2] J. Wei et al., "Shallow water bathymetry with multi-spectral satellite ocean color sensors: Leveraging temporal variation in image data," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112035.

[3] Y. Liu et al., "A downscaled bathymetric mapping approach combining multitemporal Landsat-8 and high spatial resolution imagery: Demonstrations from clear to turbid waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 65–81, Oct. 2021.

[4] B. Ai et al., "Convolutional neural network to retrieve water depth in marine shallow water area from remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2888–2898, 2020.

[5] C. J. Legleiter and B. T. Overstreet, "Mapping gravel bed river bathymetry from space," *J. Geophys. Res., Earth Surf.*, vol. 117, no. F4, p. F04024, Dec. 2012.

[6] C. J. Legleiter, M. Tedesco, L. C. Smith, A. E. Behar, and B. T. Overstreet, "Mapping the bathymetry of supraglacial lakes and streams on the Greenland ice sheet using field measurements and high-resolution satellite images," *Cryosphere*, vol. 8, no. 1, pp. 215–228, Feb. 2014.

[7] M. S. Moussavi et al., "Derivation and validation of supraglacial lake volumes on the Greenland ice sheet from high-resolution satellite imagery," *Remote Sens. Environ.*, vol. 183, pp. 294–303, Sep. 2016.

[8] Y. Ma et al., "Satellite-derived bathymetry using the ICESat-2 lidar and Sentinel-2 imagery datasets," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112047.

[9] H.-J. Hsu et al., "A semi-empirical scheme for bathymetric mapping in shallow water by ICESat-2 and Sentinel-2: A case study in the South China Sea," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 1–19, Aug. 2021.

[10] A. Albright and C. Glennie, "Nearshore bathymetry from fusion of Sentinel-2 and ICESat-2 observations," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 900–904, May 2021.

[11] N. Xu, X. Ma, Y. Ma, P. Zhao, J. Yang, and X. H. Wang, "Deriving highly accurate shallow water bathymetry from Sentinel-2 and ICESat-2 datasets by a multitemporal stacking method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6677–6685, 2021.

[12] B. Chen, Y. Yang, D. Xu, and E. Huang, "A dual band algorithm for shallow water depth retrieval from high spatial resolution imagery with no ground truth," *ISPRS J. Photogramm. Remote Sens.*, vol. 151, pp. 1–13, May 2019.

[13] A. Misra, Z. Vojinovic, B. Ramakrishnan, A. Luijendijk, and R. Ranasinghe, "Shallow water bathymetry mapping using support vector machine (SVM) technique and multispectral imagery," *Int. J. Remote Sens.*, vol. 39, no. 13, pp. 4431–4450, Jul. 2018.

[14] Z. Lee et al., "Confidence measure of the shallow-water bathymetry map obtained through the fusion of lidar and multiband image data," *J. Remote Sens.*, vol. 2021, pp. 1–16, Apr. 2021.

[15] Z. Lee, K. L. Carder, C. D. Mobley, R. G. Steward, and J. S. Patch, "Hyperspectral remote sensing for shallow waters. I. A semianalytical model," *Appl. Opt.*, vol. 37, no. 27, pp. 6329–6338, 1998.

[16] Z. Lee, K. L. Carder, C. D. Mobley, R. G. Steward, and J. S. Patch, "Hyperspectral remote sensing for shallow waters: 2. Deriving bottom depths and water properties by optimization," *Appl. Opt.*, vol. 38, no. 18, pp. 3831–3843, 1999.

[17] S. Jay and M. Guillaume, "A novel maximum likelihood based method for mapping depth and water quality from hyperspectral remote-sensing data," *Remote Sens. Environ.*, vol. 147, pp. 121–132, May 2014.

[18] S. Jay et al., "Hyperspectral remote sensing of shallow waters: Considering environmental noise and bottom intra-class variability for modeling and inversion of water reflectance," *Remote Sens. Environ.*, vol. 200, pp. 352–367, Oct. 2017.

[19] A. Minghelli-Roman and C. Dupouy, "Influence of water column chlorophyll concentration on bathymetric estimations in the lagoon of new caledonia, using several MERIS images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 739–745, Apr. 2013.

[20] J. A. Goodman, Z. Lee, and S. L. Ustin, "Influence of atmospheric and sea-surface corrections on retrieval of bottom depth and reflectance using a semi-analytical model: A case study in Kaneohe Bay, Hawaii," *Appl. Opt.*, vol. 47, no. 28, pp. F1–F11, 2008.

[21] A. Pope et al., "Estimating supraglacial lake depth in West Greenland using Landsat 8 and comparison with other multispectral methods," *Cryosphere*, vol. 10, no. 1, pp. 15–27, 2016.

[22] C. J. Legleiter, D. A. Roberts, and R. L. Lawrence, "Spectrally based remote sensing of river bathymetry," *Earth Surf. Process. Landforms*, vol. 34, no. 8, pp. 1039–1059, Jun. 2009.

[23] C. J. Legleiter, D. A. Roberts, W. A. Marcus, and M. A. Fonstad, "Passive optical remote sensing of river channel morphology and in-stream habitat: Physical basis and feasibility," *Remote Sens. Environ.*, vol. 93, no. 4, pp. 493–510, Dec. 2004.

[24] R. P. Stumpf, K. Holderied, and M. Sinclair, "Determination of water depth with high-resolution satellite imagery over variable bottom types," *Limnol. Oceanogr.*, vol. 48, pp. 547–556, Jan. 2003.

[25] C. J. Legleiter and L. R. Harrison, "Remote sensing of river bathymetry: Evaluating a range of sensors, platforms, and algorithms on the upper Sacramento river, California, USA," *Water Resour. Res.*, vol. 55, no. 3, pp. 2142–2169, Mar. 2019.

[26] L. C. Smith et al., "Efficient meltwater drainage through supraglacial streams and rivers on the southwest Greenland ice sheet," *Proc. Nat. Acad. Sci. USA*, vol. 112, no. 4, pp. 1001–1006, 2015.

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[27] H. Su, H. Liu, L. Wang, A. M. Filippi, W. D. Heyman, and R. A. Beck, "Geographically adaptive inversion model for improving bathymetric retrieval from satellite multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 465–476, Jan. 2014.
[28] C. J. Legleiter and P. J. Kinzel, "Improving remotely sensed river bathymetry by image-averaging," *Water Resour. Res.*, vol. 57, no. 3, 2021, Art. no. e2020WR028795.
[29] T. Sagawa, Y. Yamashita, T. Okumura, and T. Yamanokuchi, "Satellite derived bathymetry using machine learning and multi-temporal satellite images," *Remote Sens.*, vol. 11, no. 10, p. 1155, May 2019.
[30] S. Liu, L. Wang, H. Liu, H. Su, X. Li, and W. Zheng, "Deriving bathymetry from optical images with a localized neural network algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5334–5342, Sep. 2018.
[31] W. Lai, Z. Lee, J. Wang, Y. Wang, R. Garcia, and H. Zhang, "A portable algorithm to retrieve bottom depth of optically shallow waters from top-of-atmosphere measurements," *J. Remote Sens.*, vol. 2022, Feb. 2022, Art. no. 9831947.
[32] S. Jay, M. Guillaume, and J. Blanc-Talon, "Underwater target detection with hyperspectral data: Solutions for both known and unknown water quality," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 4, pp. 1213–1221, Aug. 2012.
[33] H. Chen et al., "Stacked spectral feature space patch: An advanced spectral representation for precise crop classification based on convolutional neural network," *Crop J.*, vol. 10, no. 5, pp. 1460–1469, Oct. 2022.
[34] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[35] W. Huang, Y. Huang, Z. Wu, J. Yin, and Q. Chen, "A multi-kernel mode using a local binary pattern and random patch convolution for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4607–4620, 2021.
[36] N. He et al., "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.
[37] Q. Xu et al., "Auto-adaptive multi-level seafloor recognition and land sea classification (AMSRLC) in reef-island zones using ICESat-2 laser altimetry," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 309–314, Sep. 2022.
[38] C. E. Parrish, L. A. Magruder, A. L. Neuenschwander, N. Forfinski-Sarkozi, M. Alonzo, and M. Jasinski, "Validation of ICESat-2 ATLAS bathymetry and analysis of ATLAS's bathymetric mapping performance," *Remote Sens.*, vol. 11, no. 14, p. 1634, 2019.
[39] T. Neumann et al., "Ice, cloud, and land elevation satellite-2 (ICESat-2) project: Algorithm theoretical basis document (ATBD) for global geolocated photons (ATL03)," 2021. [Online]. Available: https://nsidc.org/sites/nsidc.org/files/technical-references/ICESat2_ATL03_ATBD_r005.pdf
[40] L. Carrère and F. Lyard, "Modeling the barotropic response of the global ocean to atmospheric wind and pressure forcing–comparisons with observations," *Geophys. Res. Lett.*, vol. 30, no. 6, p. 1275, Mar. 2003.
[41] D. Stammer, "Accuracy assessment of global barotropic ocean tide models," *Rev. Geophys.*, vol. 52, no. 3, pp. 243–282, 2014.
[42] M. Zhao et al., "The coral communities of Yongle atoll: Status, threats and conservation significance for coral reefs in South China Sea," *Mar. Freshwater Res.*, vol. 67, no. 12, pp. 1888–1896, 2016.
[43] B.-C. Gao, "NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sens. Environ.*, vol. 58, no. 3, pp. 257–266, 1996.
[44] C. Legleiter, B. Overstreet, and P. Kinzel, "Sampling strategies to improve passive optical remote sensing of river bathymetry," *Remote Sens.*, vol. 10, no. 6, p. 935, Jun. 2018.
[45] C. Legleiter and R. Fosness, "Defining the limits of spectrally based bathymetric mapping on a large river," *Remote Sens.*, vol. 11, no. 6, p. 665, Mar. 2019.
[46] J. Li et al., "Adaptive bathymetry estimation for shallow coastal waters using Planet Dove satellites," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111302.
[47] M. Reichstein et al., "Deep learning and process understanding for data-driven Earth system science," *Nature*, vol. 566, pp. 195–204, Feb. 2019.
[48] Z. Sun et al., "A review of Earth artificial intelligence," *Comput. Geosci.*, vol. 159, Feb. 2022, Art. no. 105034.
[49] I. Caballero, R. Stumpf, and A. Meredith, "Preliminary assessment of turbidity and chlorophyll impact on bathymetry derived from Sentinel-2A and Sentinel-3A satellites in South Florida," *Remote Sens.*, vol. 11, no. 6, p. 645, Mar. 2019.
[50] Q. Vanhellmont and K. Ruddick, "Acolite for Sentinel-2: Aquatic applications of MSI imagery," in *Proc. ESA Living Planet Symp.*, vol. 740, 2016, p. 55.
[51] B. Nechad, K. G. Ruddick, and G. Neukermans, "Calibration and validation of a generic multisensor algorithm for mapping of turbidity in coastal waters," *Proc. SPIE*, vol. 7473, Sep. 2009, Art. no. 74730H.
[52] D. Zhang, Y. Chen, Y. Le, Y. Dong, G. Dai, and L. Wang, "Refraction and coordinate correction with the JONSWAP model for ICESat-2 bathymetry," *ISPRS J. Photogramm. Remote Sens.*, vol. 186, pp. 285–300, Apr. 2022.

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